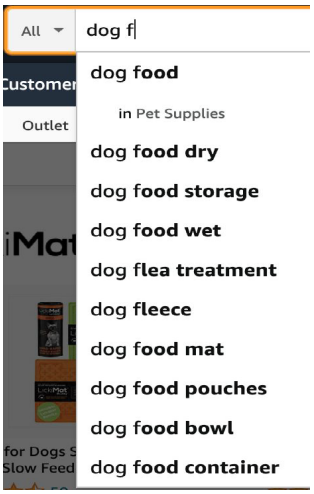
Not pleased with the results from the various supermarkets, I headed to Amazon, the popular online shop, where I encountered a *good* search engine. The second I started typing *dog*, a dropdown showed me suggestions (see the figure next to this paragraph). By default, an initial search on Amazon returns relevant (pretty close to what we're searching for) results. Using the Sort by Featured option, we can change the sort order (low to high prices or vice versa, etc.) if need be. Once the initial search is carried out, we can also drill down into other categories by selecting the department, average customer review, price range, and so on. I even tried a wrong spelling: *dig food*. Amazon asked me if I meant *dog food*. Clever, eh?



In the current digital world, search is in the spotlight. Organizations are adopting search without a second thought because they understand the business value a search engine offers and the varied use cases it can address. In the next section, we explore the exponential growth of search engines and how technology has enabled the creation of cutting-edge search solutions.

1.2 Search is the new normal

With the exponential growth of data (terabytes to petabytes to exabytes), the need for a tool that enables successful searches in needle-in-a-haystack situations is imperative. What was once touted as a simple search is now necessary functionality for most organizations' survival toolkits. Organizations are expected to provide a search function by default so customers can type in a search bar or navigate search drilldowns to locate what they need in a jiffy.

It is increasingly difficult to find websites and applications that don't have a humble search bar with a little magnifying glass. Providing full-fledged search is a competitive advantage.

Today, modern search engines strive for speed and relevancy as well as providing advanced functionality packaged in a rich set of business and technical features. Elasticsearch is one such modern search engine, embracing search and analytics with speed and performance at its heart. When dealing with search engines such as Elasticsearch, you'll come across data and search variants: structured and unstructured data and their respective searches. It is important to be familiar with these types of data in order to understand the search landscape. We look at them briefly in the next section.

1.2.1 Structured vs. unstructured (full-text) data

Data predominantly comes in two flavors: structured and unstructured. The fundamental differentiator between these two categories is the way the data is stored and analyzed. Structured data follows a predefined schema/model, while unstructured data is free-form, unorganized, and schema-free.